10.5 Solids of Revolution

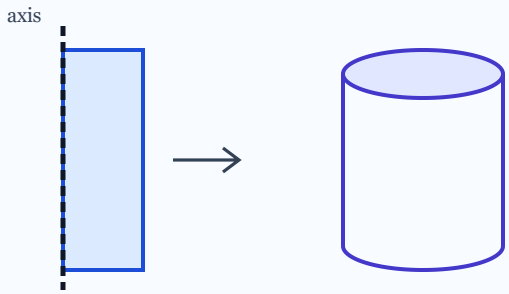
Spinning a flat 2D region a full $360°$ around a line (the **axis of revolution**) sweeps out a 3D **solid of revolution**. The three building blocks are: a rectangle about one of its sides sweeps a **cylinder**; a right triangle about a leg sweeps a **cone**; and a semicircle about its diameter sweeps a **sphere**. More advanced regions build more advanced solids: a right trapezoid about its perpendicular leg sweeps a **frustum** (a cone with the tip cut off), a rectangle that does *not* touch the axis sweeps a **hollow cylinder (tube)** — found by subtracting an inner cylinder from an outer one — and a region offset entirely from the axis sweeps a **torus** (donut shape). This worksheet identifies the solid *and* computes its volume.

Examples

Example 1: Square about a side

A 5×5 square is rotated $360°$ about one of its sides. Identify the solid and find its volume.

Solution



A rectangle (square is a special rectangle) about a side sweeps a cylinder.

The side of rotation becomes the axis; the opposite side sweeps out a circle of radius 5. Since both dimensions are 5, this is a cylinder with $r = 5$, $h = 5$.

$V = \pi r^2 h = \pi(25)(5) = 125\pi$.

Answer: cylinder; $V = 125\pi$.

Example 2: Rectangle about a side

A 4×6 rectangle is rotated $360°$ about its side of length 6. Identify the solid and find its volume.

Solution

The side of length 6 is the axis (becomes the height); the perpendicular side of length 4 sweeps out the radius. Cylinder with $r = 4$, $h = 6$.

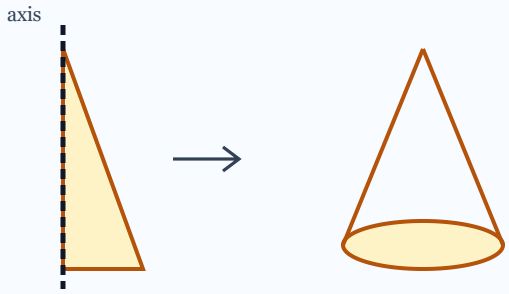
$V = \pi r^2 h = \pi(16)(6) = 96\pi$.

Answer: cylinder; $V = 96\pi$.

Example 3: Right triangle about a leg

A right triangle with legs 3 and 4 is rotated about the leg of length 4. Identify the solid and find its volume.

Solution



A right triangle about a leg sweeps a cone: the other leg becomes the radius, the axis leg becomes the height.

The leg of length 4 is the axis (height); the leg of length 3 sweeps out the radius. Cone with $r = 3$, $h = 4$.

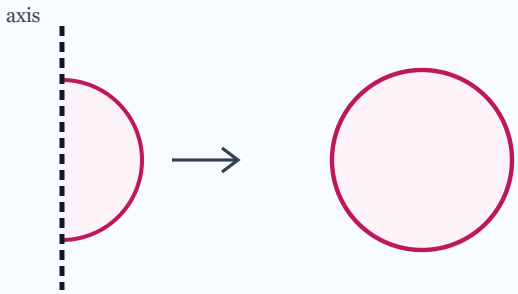
$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(9)(4) = \frac{36\pi}{3} = 12\pi.$$

Answer: cone; $V = 12\pi$.

Example 4: Semicircle about its diameter

A semicircle of radius 5 is rotated about its diameter. Identify the solid and find its volume.

Solution



A semicircle about its diameter sweeps a full sphere.

Every point on the semicircle is a distance 5 from the center of the diameter, so the swept solid is a sphere of radius 5.

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(125) = \frac{500\pi}{3}.$$

Answer: sphere; $V = \dfrac{500\pi}{3}$.

Example 5: Same rectangle, different axis

The same 4×6 rectangle from Example 2 is instead rotated about its side of length 4. Find the new volume.

Solution

Now the side of length 4 is the axis (height), and the side of length 6 sweeps the radius — the roles of r and h swap. Cylinder with $r = 6$, $h = 4$.

$$V = \pi r^2 h = \pi(36)(4) = 144\pi.$$

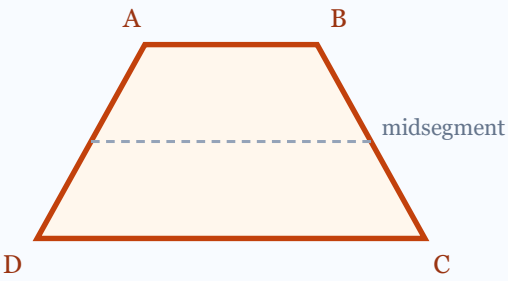
Answer: cylinder; $V = 144\pi$. *Notice this is a different volume from Example 2 (96π) — the choice of axis matters!*

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Example 6: Right trapezoid about its perpendicular leg — a frustum

A right trapezoid has parallel sides 6 and 2 and a perpendicular leg (height) of 9. It is rotated about that perpendicular leg. Identify the solid and find its volume.

Solution



A right trapezoid's two parallel sides become the two radii of a frustum when rotated about the perpendicular leg.

The perpendicular leg is the axis; the two parallel sides sweep out two different circles — a **frustum** (cone with the tip sliced off) with radii $R = 6$, $r = 2$, and height $h = 9$.

Frustum volume: $V = \frac{1}{3}\pi h(R^2 + Rr + r^2) = \frac{1}{3}\pi(9)(36 + 12 + 4) = \frac{1}{3}\pi(9)(52)$

$V = 3\pi(52) = 156\pi$.

Answer: frustum; $V = 156\pi$.

Example 7: Rectangle offset from the axis — a hollow cylinder

A rectangle spans from 3 to 5 units away from the axis (so it does not touch the axis) and is 10 units tall. It is rotated 360°. Identify the solid and find its volume.

Solution

Since the rectangle never touches the axis, it sweeps out a **hollow cylinder (tube)**: an outer cylinder of radius $R = 5$ minus an inner cylinder of radius $r = 3$, both height 10.

$V = \pi R^2 h - \pi r^2 h = \pi h(R^2 - r^2) = \pi(10)(25 - 9) = \pi(10)(16)$

$V = 160\pi$.

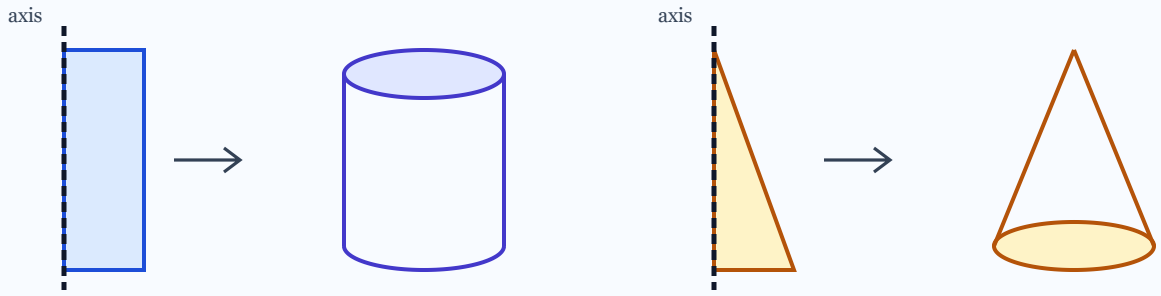
Answer: hollow cylinder; $V = 160\pi$.

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Example 8: Composite region — cylinder + cone

A rectangle (4×6, axis along the side of length 6) sits directly below a right triangle (legs 4 and 5, axis along the leg of length 5), sharing the same radius 4 and the same axis. Both are rotated together. Find the total volume of the resulting solid.

Solution



Stacking a rectangle and a right triangle on the same axis, then revolving, produces a cylinder topped with a cone.

Cylinder part: $r = 4, h = 6$: $V_1 = \pi(16)(6) = 96\pi$.

Cone part: $r = 4, h = 5$: $V_2 = \frac{1}{3}\pi(16)(5) = \frac{80\pi}{3}$.

Total: $V = 96\pi + \frac{80\pi}{3} = \frac{288\pi}{3} + \frac{80\pi}{3} = \frac{368\pi}{3}$.

Answer: $V = \frac{368\pi}{3}$.

Example 9: Right triangle about its hypotenuse — a bicone

A right triangle with legs 6 and 8 (hypotenuse 10) is rotated about its *hypotenuse*. Identify the solid and find its volume.

Solution

Rotating about the hypotenuse (instead of a leg) sweeps out two cones joined base-to-base — a **bicone** — because the altitude from the right angle to the hypotenuse becomes the shared radius, and the two segments of the hypotenuse become the two heights.

Shared radius (altitude to the hypotenuse): $r = \dfrac{(\text{leg}_1)(\text{leg}_2)}{\text{hyp}} = \dfrac{6 \times 8}{10} = 4.8.$

Heights (the two segments the altitude divides the hypotenuse into): $h_1 + h_2 = 10$ (they sum to the full hypotenuse — no need to find each one separately since the volume formula only needs the total).

$$V = \frac{1}{3}\pi r^2(h_1 + h_2) = \frac{1}{3}\pi(4.8)^2(10) = \frac{1}{3}\pi(23.04)(10) = 76.8\pi = \frac{384\pi}{5}.$$

Answer: bicone; $V = \dfrac{384\pi}{5} = 76.8\pi.$

Example 10: Region offset entirely from the axis — a torus

A semicircle of radius 3 is centered 10 units from the axis of revolution (the semicircle never touches the axis). It is rotated $360°$. Identify the solid and find its volume.

Solution

Because the region is fully offset from the axis, this sweeps out a **torus** (donut shape), not a sphere. Its volume formula is $V = 2\pi^2 R r^2$, where R is the distance from the axis to the tube's center and r is the tube's (circle's) radius.

$$V = 2\pi^2(10)(3)^2 = 2\pi^2(10)(9) = 180\pi^2.$$

Answer: torus; $V = 180\pi^2.$

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Square

Square side 7 rotated about one side. Identify the solid and find its volume.

Answer: **Verification:**

Problem 2: Right triangle

Legs 5 and 12; rotate about leg 12. Identify the solid and find its volume.

Answer: **Verification:**

Problem 3: Semicircle

Semicircle $r = 6$ about its diameter. Identify the solid and find its volume.

Answer: **Verification:**

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Problem 4: Rectangle (axis 1)

3×8 rectangle rotated about the side of length 3. Find the volume.

Answer: **Verification:**

Problem 5: Rectangle (axis 2)

The same 3×8 rectangle rotated about the side of length 8 instead. Find the volume.

Answer: **Verification:**

Problem 6: Frustum

Right trapezoid, parallel sides 6 and 2, perpendicular leg 9, rotated about that leg. Find the volume.

Answer: **Verification:**

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Problem 7: Hollow cylinder

A rectangle spans from 2 to 6 units from the axis, height 12, rotated 360°. Find the volume.

Answer: **Verification:**

Problem 8: Composite

A cylinder ($r = 3$, $h = 5$) is topped by a cone ($r = 3$, $h = 4$) on the same axis. Find the total volume.

Answer: **Verification:**

Problem 9: Identify

Rectangle rotated about one side. Name the solid.

Answer: **Verification:**

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Problem 10: Identify

Right triangle rotated about a leg. Name the solid.

Answer: **Verification:**

Problem 11: Identify

Semicircle rotated about its diameter. Name the solid.

Answer: **Verification:**

Problem 12: Identify

Right trapezoid rotated about its perpendicular leg. Name the solid.

Answer: **Verification:**

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Problem 13: Identify

Rectangle rotated with a gap between it and the axis. Name the solid.

Answer: **Verification:**

Problem 14: Word problem

A vase's outer profile is modeled as a frustum: top radius 3, bottom radius 6, height 10. Find its volume.

Answer: **Verification:**

Problem 15: Challenge

A right triangle with legs 9 and 12 (hypotenuse 15) is rotated about its hypotenuse. Find the volume of the resulting bicone.

Answer: **Verification:**

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✓ Answer Key

1. Cylinder; $V = 343\pi$
2. Cone; $V = 100\pi$
3. Sphere; $V = 288\pi$
4. 192π
5. 72π
6. 156π
7. 384π
8. 57π
9. Cylinder
10. Cone
11. Sphere
12. Frustum
13. Hollow cylinder (tube)
14. 210π
15. $\frac{1296\pi}{5}$ ($=259.2\pi$)

Great work — keep practising!